

Lab 2

Basic syntax, In/Outputs, Data Types & Arithmetic Operators in C++

Learning Objectives

The following types will be covered in this lab session

- Basic syntax of C++
- Input and output using cin and cout
- Different data types
- Memory size and address of variables
- Arithmetic operations
- Temperature conversion

1. Basic Syntax of C++

The basic syntax of C++ defines the structure and rules of writing a C++ program. Every C++ program starts with the main() function because program execution begins from it.

A C++ program usually contains:

- Header files
- Namespace declaration
- Main function
- Statements and variables

Syntax:

```
#include <iostream>

using namespace std;

int main() {

    // statements

    return 0;

}
```

Example:

```
#include <iostream>    // Header file for input and output

using namespace std;  // Standard namespace

int main() {          // Main function starts

    cout << "Welcome to C++ Programming"; // Display message

    return 0;         // End of program

} return 0; }
```

2. Input and Output in C++

C++ uses:

- "cout" For output
- "cin" for input

Both are defined in the `<iostream>` library.

- "cout" sends data to the screen
- "cin" takes data from the user through keyboard

Syntax:

```
cout << variable;
```

```
cin >> variable;
```

Example:

```
#include <iostream>  
using namespace std;  
int main() {  
    int age; // Variable declaration  
    cout << "Enter your age: "; // Ask user for input  
    cin >> age; // Take input from user  
    cout << "Your age is: " << age; // Display output  
    return 0;}
```

3. Data Types in C++

if Statement:

Data types specify the type of data a variable can store. Different data types use different memory sizes.

Common data types in C++ are:

- `int` → stores integers
- `float` → stores decimal numbers
- `double` → stores large decimal numbers
- `char` → stores single character
- `string` → stores text
- `bool` → stores true or false

Syntax:

```
int number;      //Integer data type  
float marks;    //Decimal data type  
char grade;     //Character data type
```

Example:

```
#include <iostream>  
using namespace std;  
int main() {  
    int age = 20;      // Integer type  
    float marks = 85.5; // Float type  
    char grade = 'A';  // Character type  
    cout << age << endl;  
    cout << marks << endl;  
    cout << grade << endl;  
    return 0; }
```

4. Arithmetic Operators in C++

Arithmetic operators are used to perform mathematical calculations in C++.

Common arithmetic operators are:

- $+$ $\rightarrow$ Addition
- $-$ $\rightarrow$ Subtraction
- $*$ $\rightarrow$ Multiplication
- $/$ $\rightarrow$ Division
- $\%$ $\rightarrow$ Modulus (remainder)

Syntax:

$a + b$

$a - b$

$a * b$

a / b

$a \% b$

Example:

```
#include <iostream>

using namespace std;

int main() {

    int a = 10, b = 3;    // Variable declaration

    cout << "Addition: " << a + b << endl;
    cout << "Subtraction: " << a - b << endl;
    cout << "Multiplication: " << a * b << endl;
    cout << "Division: " << a / b << endl;
    cout << "Remainder: " << a % b << endl;

    return 0;

}
```

5. Output Example

Addition: 13

Subtraction: 7

Multiplication: 30

Division: 3

Remainder: 1

In this lab, we learned the basic structure of a C++ program, input/output operations, different data types, and arithmetic operators. These concepts form the foundation of C++ programming and are essential for writing basic programs and solving mathematical problems.

Lab 2: Tasks

Task 1: Declare Different Types of Variables & Find Their Memory Size, Location, & Values

```
#include <iostream>    // Input/output library
using namespace std;   // Standard namespace
int main() {
    int num = 10;       // Integer variable
    float marks = 85.5; // Float variable
    char grade = 'A';   // Character variable
    double pi = 3.14159; // Double variable
    // Display values
    cout << "Integer value: " << num << endl;
    cout << "Float value: " << marks << endl;
    cout << "Character value: " << grade << endl;
    cout << "Double value: " << pi << endl;
    // Display memory size using sizeof operator
    cout << "\nMemory size of int: " << sizeof(num) << " bytes\n";
    cout << "Memory size of float: " << sizeof(marks) << " bytes\n";
    cout << "Memory size of char: " << sizeof(grade) << " bytes\n";
    cout << "Memory size of double: " << sizeof(pi) << " bytes\n";
    // Display memory locations (addresses)
    cout << "\nAddress of int variable: " << &num << endl;
    cout << "Address of float variable: " << &marks << endl;
    cout << "Address of char variable: " << (void*)&grade << endl;
    cout << "Address of double variable: " << &pi << endl;
    return 0; }        // End program
```

Task 2: Declare and Display a Floating-Point Variable Called “pi” and assign the value 3.14

```
#include <iostream>    // Input/output library

using namespace std;    // Standard namespace

int main() {

    float pi = 3.14;    // Declare floating-point variable

    cout << "Value of pi is: " << pi << endl; // Display value

    return 0;          // End program

}
```


Task 3: Convert Temperature from Celsius to Fahrenheit (Hint: $F = (C \times 9/5) + 32$)

```
#include <iostream>    // Input/output library

using namespace std;    // Standard namespace

int main() {

    float celsius;      // Variable for Celsius temperature

    float fahrenheit;   // Variable for Fahrenheit temperature

    cout << "Enter temperature in Celsius: ";

    cin >> celsius;     // Take Celsius input

    // Apply conversion formula

    fahrenheit = (celsius * 9 / 5) + 32;

    // Display result

    cout << "Temperature in Fahrenheit: " << fahrenheit << endl;

    return 0;          // End program

}
```